

# Sydney Girls High School



## MATHEMATICS

### YEAR 11 Preliminary Course

### ASSESSMENT TASK 3

### 2017

Reading time : 5 minutes

Writing time : 60 minutes

Total marks : 60

Topics: Basic Arithmetic and Algebra, Plane Geometry, Real Functions,  
Trigonometric Ratios, The Linear Function,  
Introductory Calculus: Tangent to a curve, differentiation of a function.

#### General Instructions:

- There are four (4) questions which are of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each question on a new page.
- Write on one side of the paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number clearly at the top of each question and clearly number each question.
- Write using a black or blue pen.

Student Name: \_\_\_\_\_ Teacher Name: \_\_\_\_\_

**QUESTION 1****MARKS**

a) Evaluate  $(8.315 \times 10^3) \div (2.5 \times 10^{-5})$  answering in scientific notation. 1

b) Fully factorise  $4m - 4n - m^2 + mn$  1

c) Solve  $(x+2)^2 - 25 = 0$  2

d) If  $\frac{4}{2-\sqrt{3}} = a + b\sqrt{3}$  find the values of  $a$  and  $b$ . 2

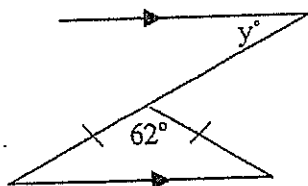
e) Given that  $\sin \theta = \frac{5}{13}$  and  $\tan \theta < 0$ , what is the exact value of  $\cos \theta$ ? 2

f) State the domain and range of the following function 2

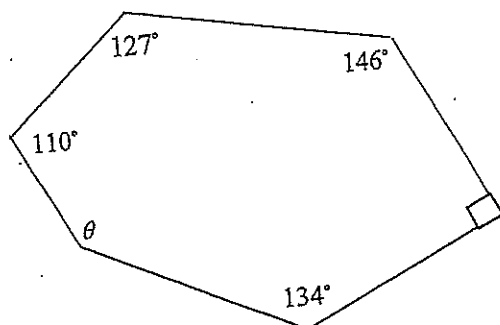
$$y = \sqrt{9 - x^2}$$

g) Find the value of the pronumeral in each case

i) 1



ii) 2



h) The curve  $y = ax^2 - 6x + 3$  has a stationary point at  $x = 1$ . 2  
What is the value of  $a$ ?

**QUESTION 2****MARKS**

a) Solve  $2^{3x-1} = \frac{1}{4}$

**2**

b) Differentiate the following:

i)  $\sqrt{1-x}$

**1**

ii)  $\frac{7x}{3-x}$

**2**

c) Given  $f(x) = \begin{cases} x+1 & x \geq 3 \\ x^2 + 2x - 1 & -1 < x < 3 \\ 3^x & x \leq -1 \end{cases}$ ,

**2**find the value of  $f(2) + f(-2) - f(6)$ .d) Find the equation of the tangent to the curve  $y = 3x^2 - x^3$  at  $x = -1$ .**3**

e) Solve  $|8 - 2m| \leq 70$

**2**

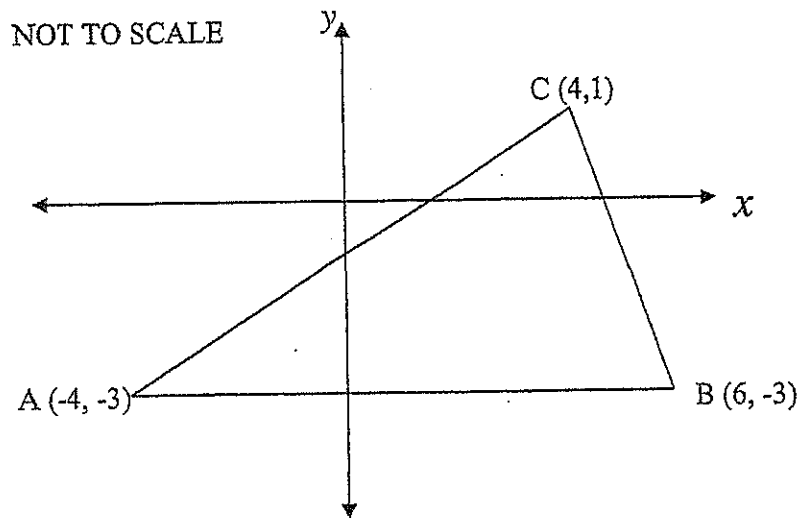
f) Show that  $\frac{(1 + \tan^2 \theta) \cot \theta}{\operatorname{cosec}^2 \theta} = \tan \theta$

**3**

## QUESTION 3

MARKS

- a) The diagram below shows  $\triangle ABC$  with the vertices  $A(-4,-3)$ ,  $B(6,-3)$  and  $C(4,1)$ .



- |      |                                                                 |   |
|------|-----------------------------------------------------------------|---|
| i)   | Find the gradient of the interval BC.                           | 1 |
| ii)  | Prove A, B and C are the vertices of a right-angled triangle.   | 1 |
| iii) | Find the distance BC in exact form.                             | 1 |
| iv)  | Find the size of $\angle CAB$ to the nearest minute.            | 2 |
| v)   | Show that the coordinates of M, the midpoint of AB are (1,-3).  | 1 |
| vi)  | Hence, show that the equation of the circle with diameter AB is | 2 |

$$x^2 + y^2 - 2x + 6y - 15 = 0$$

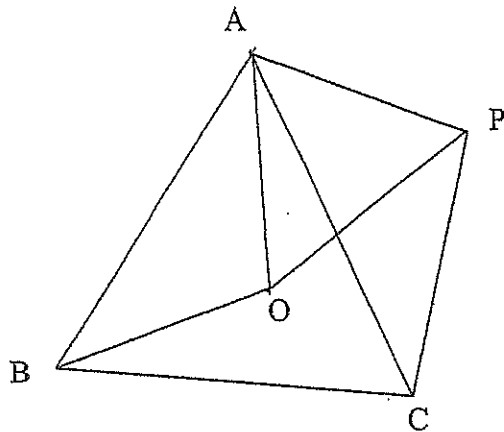
- b) Find the derivative of  $y = (3x^2 + 1)(4 - x)^4$ , expressing your answer in factorised form. 2

- c) Given  $f(x) = x^2 + 4$ ,  $g(x) = x - x^3$  and  $h(x) = f(x) \times g(x)$ ,  
show that  $h(x)$  is an odd function. 1

## QUESTION 3 (continued)

MARKS

d)



In the figure above, triangles ACB and APO are equilateral.

i) Copy this diagram onto your answer sheet and label all information given.

ii) Explain why  $\angle BAO = \angle PAC$ .

1

(Hint: Let  $\angle OAC = x$ )

iii) Prove  $\triangle AOB \equiv \triangle APC$ .

2

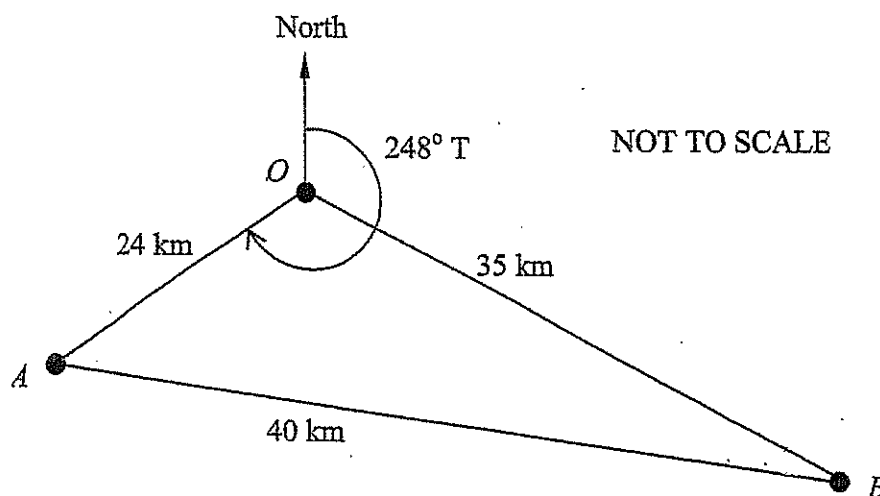
iv) Hence, prove  $OB = CP$ .

1

## QUESTION 4

## MARKS

- a) Differentiate  $y = 5x^2 + 1$  from first principles. 2
- b) Draw a neat sketch showing the region on the number plane where the inequalities  $y < x^2 - 4x + 3$  and  $y \leq x + 3$  hold simultaneously. 2
- c) Consider the curve  $y = x^3 - 12x^2 + 36x$ .
- i) Find any stationary points. 2
- ii) Determine the nature of the stationary points. 1
- d) A triangular section of rainforest is to be designated for a species count. The shape is represented below by  $\triangle AOB$ . The bearing of landmark A from landmark O is  $248^\circ$  and is 24km in distance. The distance from landmark A to B is 40km and from landmark B to O is 35km.

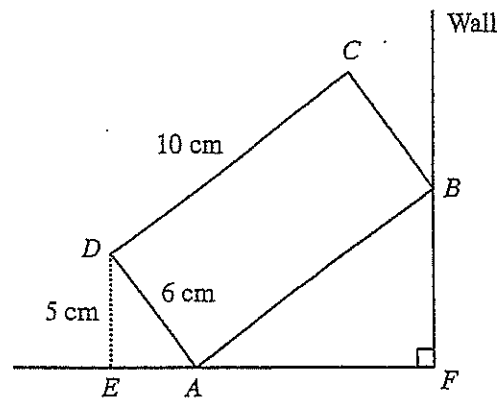


- i) Show that  $\angle AOB$  is  $83^\circ$ . 1
- ii) Hence, calculate the area of this section of the rainforest. 1
- iii) What is the bearing of landmark O from landmark B? 2

## QUESTION 4 (continued)

MARKS

e)



In the diagram above,  $ABCD$  is the cross section of a rectangular block which has been placed against a wall. The block is 10cm long and 6cm wide. The outermost edge (indicated by the point  $D$  in the diagram) of the block is 5cm above the ground.

i) Prove that  $\triangle EAD$  is similar to  $\triangle FBA$

2

ii) At what height does the block touch the wall (indicated by the point  $B$  in the diagram)?

2

Give your answer in simplest surd form.

THE END

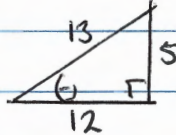
Question One Yr 11 Task 2 Adv

a)  $3.326 \times 10^8 \checkmark$  (no other answers accepted) (1)

b)  $4m - 4n - m^2 + mn$   
 $4(m-n) - m(m-n)$   
 $(m-n)(4-m) \checkmark$  (1)

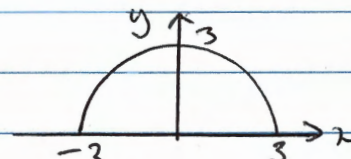
c)  $(x+2)^2 - 25 = 0$   
 $(x+2)^2 - 5^2 = 0$   
 $(x+2+5)(x+2-5) = 0$   
 $(x+7)(x-3) = 0$   
 $x = -7 \checkmark, x = 3 \checkmark$  (2)

d)  $\frac{4}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = \frac{8+4\sqrt{3}}{4-3}$   
 $= 8+4\sqrt{3}$  or  $a=8, b=4 \checkmark$  (2)

e)  $\sin \theta = \frac{5}{13}$    $\tan \theta < 0$ 

|   |   |
|---|---|
| C | A |
| + | - |

  
 $\cos \theta = -\frac{12}{13} \checkmark$  (2)

f)   $-3 \leq x \leq 3 \checkmark, 0 \leq y \leq 3 \checkmark$  (2)

g) i)  $y = \frac{180^\circ - 62^\circ}{2}$   
 $= 59^\circ$  (1)  $\checkmark$

ii)  $S_n = (n-2)180^\circ, n=6$   
 $S_n = 4 \times 180^\circ = 720^\circ \checkmark$   
 $\theta = 720^\circ - (110^\circ + 127^\circ + 146^\circ + 90^\circ + 134^\circ)$   
 $= 113^\circ \checkmark$  (2)

h)  $y = ax^2 - 6x + 3, \frac{dy}{dx} = 2ax - 6 \checkmark$   
for stg pt  $\frac{dy}{dx} = 0$  i.e.  $2a(1) - 6 = 0$   
 $2a = 6$  (2)  
 $a = 3 \checkmark$

Assessment Task 3  
Year 11 Preliminary course  
Mathematics

2 a)

$$2^{3x-1} = \frac{1}{4}$$

$$2^{3x-1} = 2^{-2}$$

$$3x-1 = -2$$

$$3x = -1$$

$$x = -\frac{1}{3}$$

majority of students got it correct.

b) i)  $f(x) = \sqrt{1-x}$

$$f'(x) = \frac{1}{2} (1-x)^{-\frac{1}{2}} (-1)$$

$$= -\frac{(1-x)^{-\frac{1}{2}}}{2}$$

$$= -\frac{1}{2\sqrt{1-x}}$$

Some students made mistakes while attempting the chain rule.  $(\frac{1}{2} (1-x)^{-\frac{1}{2}} \frac{d}{dx}(1-x))$   
so marks were deducted.

(ii)

$$f(x) = \frac{7x}{3-x}$$

This is an example of quotient rule. Some students made the mistake when attempting to sub values in the final formula.

$$u = 7x \quad u' = 7$$

$$v = 3-x \quad v' = -1$$

$$f(x) = \frac{u}{v}$$

$$f'(x) = \frac{v u' - u v'}{v^2}$$

$$f'(x) = \frac{(3-x)7 - 7x(-1)}{(3-x)^2} = \frac{21-7x+7x}{(3-x)^2}$$

$$f'(0) = \frac{21}{(3-0)^2}$$

Assessment Task-3  
Year 11 Preliminary course,  
Mathematics.

2

c)

$$f(2) = 4 + 4 - 1 = 7$$

$$f(-2) = 3^{-2} = \frac{1}{9}$$

$$f(6) = 7$$

$$f(2) + f(-2) - f(6) = 7 + \frac{1}{9} - 7 = \frac{1}{9}$$

some students were confused when choosing the correct function value for the given ~~ran~~ domain. So they were penalised.

2

d)

$$y = 3x^2 - x^3 \quad \text{at } x = -1, y = +3 + 1 = 4$$

$$y' = 6x - 3x^2$$

gradient  $m$  at  $x = -1$

$$m = -6 - 3 = -9$$

some students ~~miscaluate~~ miscalculated the gradient.

Equation of tangent

$$y - 4 = -9(x + 1)$$

$$y - 4 = -9x - 9$$

$$9x + y + 5 = 0$$

e)

$$|8 - 2m| \leq 70$$

~~ma~~ many students made the mistake in this question while processing the values.

$$-70 \leq 8 - 2m \leq 70$$

$$-78 \leq -2m \leq 62$$

$$39 \geq m \geq -31$$

$$-31 \leq m \leq 39$$

2

L.H.S

f)

$$= \frac{\sec^2 \theta \times \cot \theta}{\operatorname{cosec}^2 \theta} = \frac{1}{\cos^2 \theta} \times \frac{\cos \theta}{\sin \theta}$$

$$\frac{1}{\sin^2 \theta}$$

$$= \frac{\sin^2 \theta}{\cos \theta} \times \frac{1}{\sin \theta} = \tan \theta$$

$$= \text{R.H.S}$$

Hence proved.

Some students took a complicated and unnecessary long way to prove the above relation. Always look for the most economical way which will save you time.

### Question 3

a)

$$i) m_{BC} = \frac{-3-1}{6-4}$$

$$= \frac{-4}{2}$$

$$m_{BC} = -2$$

(1)

$$ii) m_{AC} = \frac{-3-1}{-4-4}$$

$$= \frac{-4}{-8}$$

$$= \frac{1}{2}$$

$$m_{BC} \times m_{AC} = -1 \quad (AC \perp BC)$$

(1)

$\therefore \triangle ABC$  is right-angled at  $\angle ACB$ .

$$iii) BC = \sqrt{(6-4)^2 + (-3-1)^2}$$

$$= \sqrt{4+16}$$

$$BC = 2\sqrt{5}$$

(1)

$$iv) \sin \angle CAB = \frac{BC}{AB}$$

$$\sin \angle CAB = \frac{2\sqrt{5}}{10}$$

(2)

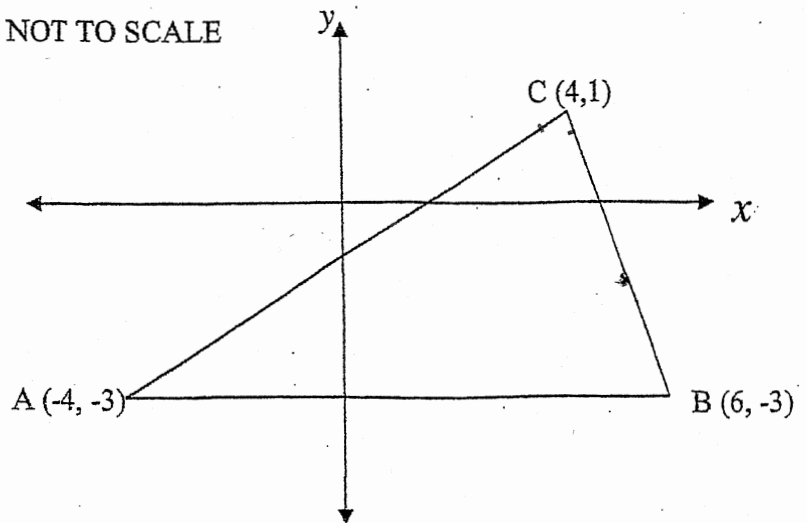
$$\therefore \angle CAB = 26^\circ 34'$$

$$v) M = \left( \frac{-4+6}{2}, \frac{-3+(-3)}{2} \right)$$

$$M = (1, -3)$$

(1)

NOT TO SCALE



vi) Equation of circle:

$$(x-a)^2 + (y-b)^2 = r^2$$

$$(x-1)^2 + (y+3)^2 = 5^2$$

(2)

$$x^2 - 2x + 1 + y^2 + 6y + 9 = 25$$

$$x^2 + y^2 - 2x + 6y - 15 = 0$$

as required

Q3

$$b) y = (3x^2 + 1)(4 - x)^4$$

$$\frac{dy}{dx} = (3x^2 + 1) \times 4(4 - x)^3 \times -1 + 6x(4 - x)^4 \checkmark$$

$$= -4(3x^2 + 1)(4 - x)^3 + 6x(4 - x)^4$$

$$= (4 - x)^3 [-12x^2 - 4 + 6x(4 - x)]$$

$$= (4 - x)^3 [-12x^2 - 4 + 24x - 6x^2]$$

$$= (4 - x)^3 (-18x^2 + 24x - 4)$$

$$= 2(4 - x)^3 (-9x^2 + 12x - 2) \checkmark \quad (2)$$

$$c) h(x) = f(x) \times g(x)$$

$$= (x^2 + 4)(x - x^3)$$

$$= x^3 - x^5 + 4x - 4x^3$$

$$= -x^5 - 3x^3 + 4x$$

$$h(-x) = -(-x)^5 - 3(-x)^3 + 4(-x)$$

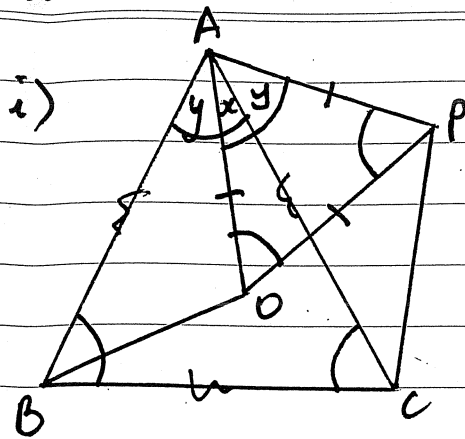
$$= x^5 + 3x^3 - 4x$$

$$= -(-x^5 - 3x^3 + 4x)$$

$$= -h(x) \quad (1)$$

$\therefore h(x)$  is odd.

Q3d).



ii)  $\angle BAC = 60^\circ$  ( $\triangle ABC$  is equilateral).

$$\angle BAO + \angle OAC = 60^\circ$$

$$\angle BAO + x = 60^\circ$$

$$\therefore \angle BAO = 60^\circ - x.$$

$\angle PAO = 60^\circ$  ( $\triangle APO$  is equilateral).

$$\angle PAC + \angle OAC = 60^\circ$$

$$\angle PAC + x = 60^\circ$$

$$\therefore \angle PAC = 60^\circ - x.$$

①

$$\therefore \angle BAO = \angle PAC.$$

iii) In  $\triangle AOB$  and  $\triangle APC$

$AB = AC$  ( $\triangle ABC$  is equilateral).

$AO = AP$  ( $\triangle APO$  is equilateral)

$\angle BAO = \angle PAC$  (as above).

$\therefore \triangle AOB \equiv \triangle APC$  (SAS)

②

iv)  $\therefore OB = CP$  (corresponding sides in congruent  $\triangle$ s).

③

# Question 4 (15 marks)

a)  $y = 5x^2 + 1$

let  $f(x) = 5x^2 + 1$

$$\begin{aligned} f(x+h) &= 5(x+h)^2 + 1 \\ &= 5(x^2 + 2xh + h^2) + 1 \\ &= 5x^2 + 10xh + 5h^2 + 1 \end{aligned}$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{5\cancel{x^2} + 10xh + 5h^2 + 1 - 5\cancel{x^2} - 1}{h} \\ &= \lim_{h \rightarrow 0} \frac{10xh + 5h^2}{h} \\ &= \lim_{h \rightarrow 0} \cancel{h} \frac{(10x + 5h)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} 10x + 5h \\ &= 10x \end{aligned}$$

(overall question was completed very well, with a few making careless errors)

$\frac{1}{2}$

(Graphs were drawn poorly and incomplete. Most found correct region.)

$$y = x^2 - 4x + 3$$

$$y = (x-3)(x-1)$$

$x = 3, x = 1$

$$y = x + 3$$

at  $x = 0, y = 3$

$x = -3, y = 0$

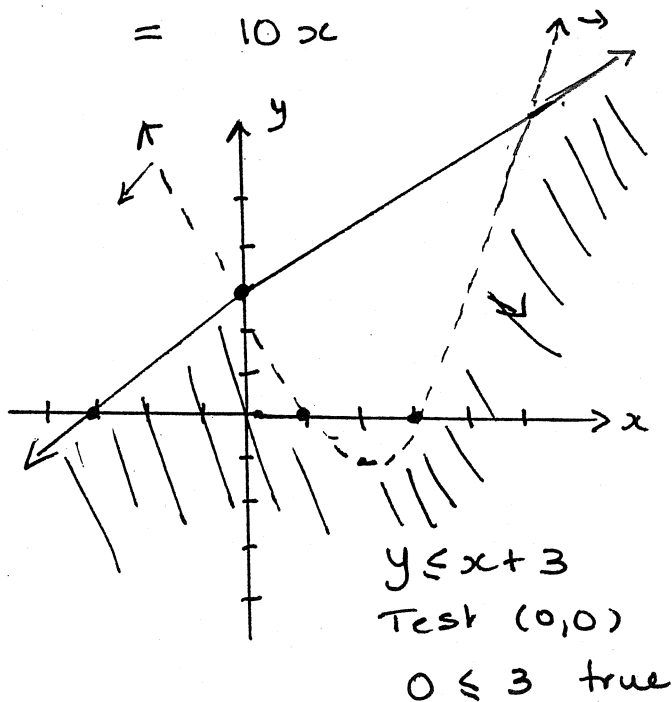
$$y < x^2 - 4x + 3$$

Test  $(2, 0)$

$0 < -1$  false

$\frac{1}{2}$

b)



4 c)  $y = x^3 - 12x^2 + 36x$

$y' = 3x^2 - 24x + 36$

i) Stationary points at  $y' = 0$

$3x^2 - 24x + 36 = 0$

$x^2 - 8x + 12 = 0$

$(x - 6)(x - 2) = 0$

$x = 6$  or  $x = 2$

$y = 0$

$P_1(6, 0)$

$y = 32$

$P_2(2, 32)$

(Most students found stationary pts)

(Full marks were given if found  $x = 6$  and  $x = 2$  without  $y$ -values however, this will not happen in the next assessment!)

Both  $x$  &  $y$  values must always be found.

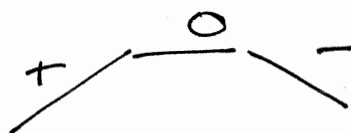
ii) Nature of stationary points

|      |   |   |   |
|------|---|---|---|
| $x$  | 5 | 6 | 7 |
| $y'$ | - | 0 | + |



Minimum

|      |   |   |   |
|------|---|---|---|
| $x$  | 1 | 2 | 3 |
| $y'$ | + | 0 | - |



Maximum

$\therefore P_1(6, 0)$  is a minimum Stationary point

$P_2(2, 32)$  is a maximum stationary point.

(Overall question answered extremely well).

(1)

$$4 d) i) \cos \angle AOB = \frac{b^2 + a^2 - o^2}{2ab}$$

$$= \frac{24^2 + 35^2 - 40^2}{2 \times 24 \times 35}$$

(Most students answered very well)

$$\cos \angle AOB = 0.11964...$$

$$\angle AOB = \cos^{-1}(0.11964...)$$

$$\therefore \angle AOB = 83^\circ$$



$$ii) \text{Area} = \frac{1}{2} ab \sin O$$

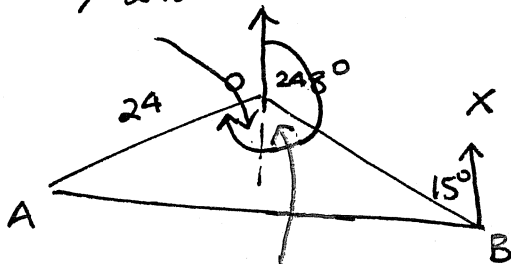
$$= \frac{1}{2} \times 24 \times 35 \times \sin 83^\circ$$

$$= 416.8693...$$

$$= 417 \text{ km}^2$$



$$iii) 248 - 180 = 68^\circ$$



$$83 - 68 = 15^\circ$$

(Most students answered extremely well)

$$\therefore \angle BOX = 15^\circ$$

(co-interior  $\angle$ 's) (1)

$\therefore$  Bearing of O from B

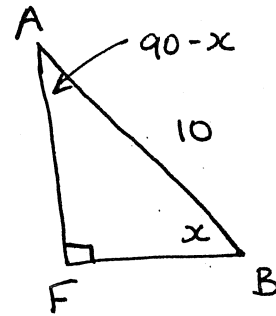
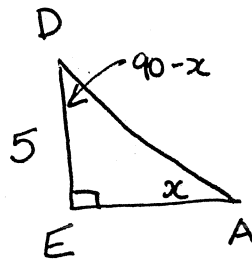
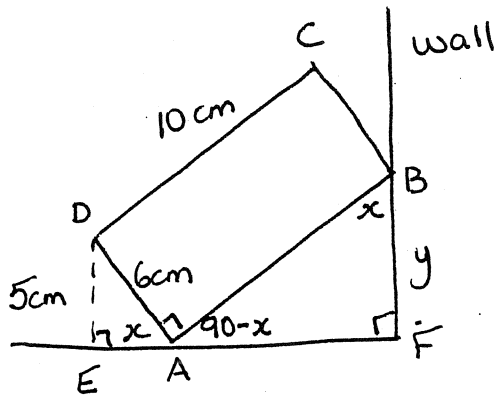
is: N  $15^\circ$  W or (1)

$$360 - 15^\circ = 345^\circ$$

(A lot of students had issues determining correct bearing).



4 e)



In  $\triangle EDA$  and  $\triangle FBA$

i) let  $x = \angle DAE$

$$\begin{aligned}\angle BAF &= 180 - (90 + x) \\ &= 180 - 90 - x \\ &= 90 - x\end{aligned}$$

(straight  $\angle$ )

(Question completed poorly)

$$\begin{aligned}\angle ABF &= 180 - 90 - (90 - x) \quad (\text{Angle sum of } \triangle) \\ &= 180 - 90 - 90 + x \\ &= x\end{aligned}$$

$$\therefore \angle DAE = \angle ABF = x$$

(as shown)

2

1 mark if stated \*

$$* \angle DEA = \angle AFB = 90^\circ \quad (\text{given})$$

$$\angle EDA = \angle FAB = 90 - x \quad (\text{angle sum of } \triangle)$$

$$* \therefore \triangle EDA \parallel \triangle FBA \quad (\text{equiangular})$$

ii) Find FB

Since  $\triangle$ 's are similar corresponding sides are in the same ratio

$$\frac{AF}{DE} = \frac{AB}{DA}$$

$$\frac{AF}{5} = \frac{10}{6}$$

$$\therefore AF = \frac{10 \times 5}{6} = \frac{25}{3}$$

$\therefore$  Block touches the wall at a height of  $\frac{5\sqrt{11}}{3}$  cm above floor

$\therefore$  In  $\triangle AFB$ ,

$$\begin{aligned}(BF)^2 &= (AB)^2 - (AF)^2 \\ &= 10^2 - \left(\frac{25}{3}\right)^2\end{aligned}$$

$$\begin{aligned}(BF)^2 &= 100 - \frac{625}{9} \\ &= \frac{275}{9}\end{aligned}$$

$$\therefore BF = \sqrt{\frac{275}{9}} = \frac{5\sqrt{11}}{3} \text{ cm}$$

2